

University admission management system

Submitted By

Student Name	Student ID
Mahtab hasan	252-15-860
Ranggon Saha Riti	252-15-787
Sneha Hazra	252-15-907
Sumaiya Zaman Anika	252-15-874
Sreedam Ghosh Sohag	252-15-838

MINI LAB PROJECT REPORT

This Report Presented in Partial Fulfillment of the course **CSE124:**
Data structure in the Computer Science and Engineering
Department



DAFFODIL INTERNATIONAL UNIVERSITY
Dhaka, Bangladesh

April 7, 2026

DECLARATION

We hereby declare that this lab project has been done by us under the supervision of **Raja Tariqul Islam Tusher, Assistant Professor**, Department of Computer Science and Engineering, Daffodil International University. We also declare that neither this project nor any part of this project has been submitted elsewhere as lab projects.

Submitted To:

Raja Tariqul Islam Tusher

Assistant Professor

Department of Computer Science and Engineering

Daffodil International University

Submitted by

Mahtab Hasan ID:252-15-860 Dept. of CSE, DIU	
Ranggon Saha Riti ID:252-15-787 Dept. of CSE, DIU	Sumaiya Zaman Anika ID:252-15-874 Dept. of CSE, DIU
Sneha Hazra ID:252-15-907 Dept. of CSE, DIU	Sreedam Ghosh Sohag ID:252-15-838 Dept. of CSE, DIU

COURSE & PROGRAM OUTCOME

The following course have course outcomes as following:.

Table 1: Course Outcome Statements

CO's	Statements
C01	Define and Relate classes, objects, members of the class, and relationships among them needed for solving specific problems
C02	Formulate knowledge of object-oriented programming and Java in problem solving
C03	Analyze Unified Modeling Language (UML) models to Present a specific problem
C04	Develop solutions for real-world complex problems applying OOP concepts while evaluating their effectiveness based on industry standards.

Table 2: Mapping of CO, PO, Blooms, KP and CEP

CO	PO	Blooms	KP	CEP
C01	P01	C1, C2	KP3	EP1,EP3
C02	P02	C2	KP3	EP1,EP3
C03	P03	C4, A1	KP3	EP1,EP2
C04	P03	C3, C6, A3, P3	KP4	EP1,EP3

The mapping justification of this table is provided in section 4.3.1, 4.3.2 and 4.3.3.

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Chapter 1

Introduction

The **University Admission Management System** is a web-based application designed to automate and streamline the traditional manual admission process. This chapter provides an overview of the project, including its objectives, the problems it aims to solve, and the overall scope of the system.

1.1 Introduction

University admissions are often manual, slow, and error-prone. This project develops a system to automate and improve the admission process. This system provides a centralized platform where prospective students can register, upload their academic credentials, choose their desired departments, and track their application status in real-time. Simultaneously, it empowers the university administration with tools to verify documents, compute merit lists based on predefined criteria, and manage seat allocations with minimal human intervention.

1.2 Motivation

The primary motivation for developing the University Admission Management System arises from the urgent need to modernize the traditional, labor-intensive enrollment process. Traditional manual systems are often plagued by systemic bottlenecks, including excessive paperwork, susceptibility to human error, and the physical constraints of geographical distance. This project is driven by the desire to implement a high-performance digital solution that ensures absolute data integrity and transparency through a centralized database.

1.3 Objectives

The primary objective of this project is to develop a reliable and efficient system that automates the university admission cycle. The specific objectives are as follows:

- **To Automate the Application Process:** Replace manual form submission with a streamlined online registration system that allows candidates to apply from anywhere.
- **To Improve Administrative Control:** Empower the university staff with a robust admin panel to monitor seat capacity, verify documents, and manage departmental allocations efficiently, reducing the risk of data loss and eliminating human errors in record-keeping.
- **To Reduce Operational Costs:** Minimize the use of physical paper and the need for large numbers of on-site staff, making the admission process more sustainable and cost-effective.
- **To Ensure Data Accuracy and Security:** Implement a centralized database to store student information securely, reducing the risk of data loss and eliminating human errors in record-keeping.

1.4 Feasibility Study

The feasibility study evaluates the viability of the **University Admission Management System** across three main dimensions:

- **Technical Feasibility:** The project is technically sound as it utilizes standard hardware and widely supported software technologies. The development team possesses the necessary skills to implement the database and user interface.
- **Economic Feasibility:** Although there is an initial development cost, the system is highly cost-effective in the long run. It eliminates expenses related to manual paperwork, physical storage, and large-scale administrative staffing.
- **Operational Feasibility:** The system is designed to be user-friendly, requiring minimal training for students and staff. It integrates seamlessly into the university's workflow by replacing slow manual queues with an efficient digital process.

1.5 Gap Analysis

Existing systems often lack speed and user-friendly design. This project will improve efficiency and usability.

1.6 Project Outcome

Admin Benefits:

1. Easily store and manage all applicant data.
2. Complete the admission process quickly and accurately.
3. Generate and monitor reports efficiently.

User (Student) Benefits:

1. Submit applications easily online.
2. Check application status quickly.
3. Save time and cost without visiting physically.

Chapter 2

Proposed Methodology/Architecture

This chapter describes the proposed methodology and overall system architecture of the University Admission Management System. It explains how the system is designed, developed, and how different components interact to ensure efficient admission processing.

2.1 Requirement Analysis & Design Specification

2.1.1 Overview

The University Admission Management System is a web-based application designed to simplify and automate the admission process. It allows students to apply online and enables administrators to manage applications, data, and selection efficiently.

2.1.2 Proposed Methodology/ System Design

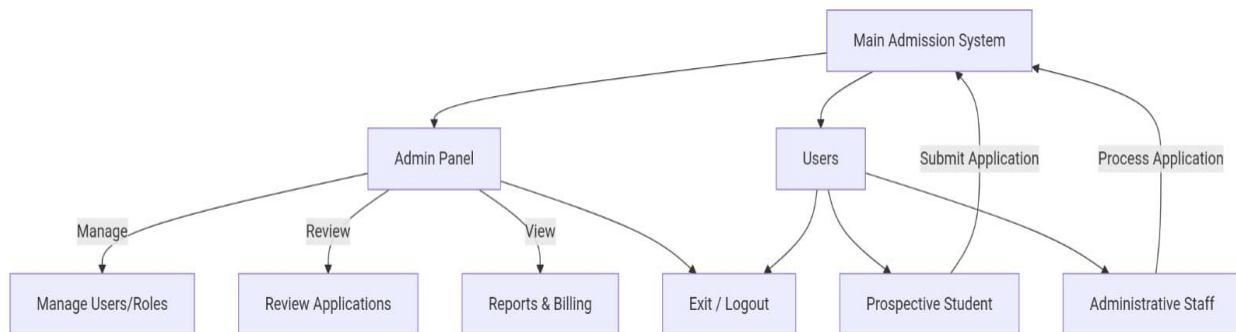


Figure 2.1: This is a sample diagram

2.1.3 UI Design

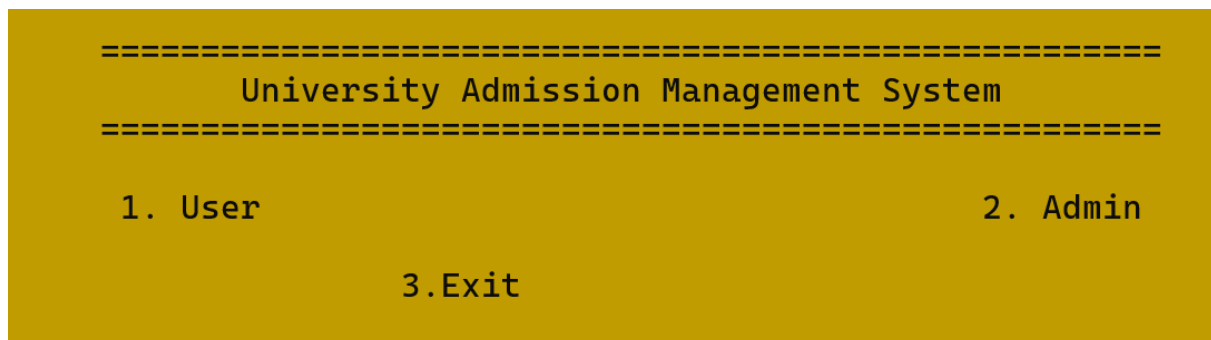


Fig: 2.1:Main manu

This UI represents a simple main menu for a University Admission Management System. It provides three options: User access, Admin access, and Exit for closing the system. The design is clean and text-based, making navigation straightforward for basic operations.

```
--- Student Dashboard ---
1. Create Account
2. Log In
3. Exit
Choice: 2
Username: anika
Password: 123
Login successful!

--- MENU ---
1. About University
2. Departments Info
3. Apply for Registration
4. Log Out
Choice: |
```

Fig: 2.2: Student Management system

This UI shows the student dashboard of the University Admission Management System after successful login.

It allows students to create accounts, log in, and access options like university info, department details, and registration.

The interface is simple and menu-driven, making it easy for users to navigate different functions.

```
Enter your choice: 2
Enter Admin Password: admin123
Admin Login Successful!

--- ADMIN PANEL ---
1. Update University Data
2. Manage Applied Students
3. Exit
Choice: 1

Update University Data:
1. Faculty Details
2. Students & Alumni Details
3. Tuition Fees
4. GPA Requirement
5. Exit
```

Fig:2.3: Admin management system

This UI represents the admin panel of the University Admission Management System. It allows the admin to manage student data, admissions, and system information. The design is simple and functional, focusing on easy control and management tasks.

2.2 Overall Project Plan

- **Planning:**
Define the project goals, scope, and timeline clearly.
Identify the main features and target users.
- **Requirement Analysis:**
Gather all functional and non-functional requirements.
Understand user needs for both admin and students.
- **System Design:**
Design the system architecture, database, and UI layout.
Plan how different modules will interact.
- **Development:**
Build the frontend and backend of the system.
Integrate database and implement all features.
- **Testing:**
Check the system for bugs and errors.
Ensure performance, security, and accuracy.
- **Deployment:**
Launch the system on a live server.
Make it accessible to users for real use.
- **Maintenance:**
Monitor system performance regularly.
Fix issues and update features when needed.

Phase	Duration	Activities
Planning	Week 1	Define goals, scope, and timeline Identify key features
Requirement Analysis	Week 2	Collect user requirements Analyze system needs
System Design	Week 3-4	Design database and architecture Create UI and system layout
Development	Week 5-7	Develop frontend and backend Implement core functionalities
Testing	Week 8	Perform testing and debugging Ensure system performance
Deployment	Week 9	Deploy system on server Make system live
Maintenance	Week 10	Monitor system and fix issues Update and improve features

Chapter 3

Implementation and Results

This chapter describes how the University Admission Management System is implemented and presents the results obtained after testing the system. It explains the working modules and evaluates system performance and outcomes.

3.1 Implementation

The system primarily utilizes **Doubly Linked Lists** to manage dynamic data. This allows for efficient insertion, deletion, and bidirectional traversal of records.

- **struct Student:** Holds comprehensive details of a student, including name, password, department, GPA, and contact information.
- **StudentNode:** A doubly linked list node that stores student data along with `prev` and `next` pointers.
- **FormNode:** A dedicated linked list to manage pending admission applications.

Student Authentication (Login & Signup)

The system allows new users to create an account using `createStudentAccount()`. The data is stored in the `studentHead` linked list. Existing users are verified using the `findStudentByNamePass()` function, which traverses the list to match credentials.

Application Processing

When a student applies for a department, the system:

1. Validates the student's GPA against department requirements (e.g., CSE needs 9.0).
2. Collects additional details (Father's name, City, Transaction ID).
3. Inserts the application into the `formHead` list using `insertFormAtEnd()`.

Admin Dashboard

The Admin Panel is protected by a password (`admin123`). It provides high-level control, including:

- **University Data Management:** Viewing and updating department-wise faculty, fees, and GPA requirements.
- **Application Management:** Searching, counting, or deleting student applications from the linked list.

Dynamic memory allocation (`malloc`) is used to create new nodes for students and forms. This ensures the system only uses as much RAM as necessary for the current number of records.

3.2 Performance Analysis

This chapter evaluates the performance of the University Admission Management System in terms of efficiency, accuracy, and usability.

- **Time Efficiency:**

Faster than manual system; operations like search and insert are quick.

- **Space Efficiency:**

Uses dynamic memory (linked list), so memory is used efficiently.

- **Accuracy:**

Reduces human errors in data entry and processing.

- **Response Time:**

Immediate output for login, form submission, and search.

- **Scalability:**

Can handle more users by extending linked list.

- **Reliability:**

System runs smoothly with minimal failure.

- **Limitations:**

- Linear search is slower for large data
- Console-based (no GUI)

3.3 Results and Discussion

This chapter presents the outcomes of the implemented system and discusses its performance, advantages, and limitations based on testing.

```
--- Student Dashboard ---
1. Create Account
2. Log In
3. Exit
Choice: 2
Username: anika
Password: 123
Login successful!

--- MENU ---
1. About University
2. Departments Info
3. Apply for Registration
4. Log Out
```

Fig 3.1 : user interface

```
--- ADMIN PANEL ---
1. Update University Data
2. Manage Applied Students
3. Exit
Choice: 1

Update University Data:
1. Faculty Details
2. Students & Alumni Details
3. Tuition Fees
4. GPA Requirement
5. Exit
```

Fig 3.2 : Admin interface

Results

- Successful **student registration and login**

- Smooth **admission form submission**
- Correct **GPA-based department selection**
- Efficient **admin management of applications**
- Accurate **display of student information**

Discussion

- The system is **simple and user-friendly**
- Reduces **manual work and time**
- Provides **organized data management** using linked list
- Works well for **small to medium data size**
- Limitation:
 - Uses **linear search (slower for large data)**
 - No graphical user interface (GUI)

Chapter 4

Engineering Standards and Mapping

This chapter describes the engineering standards followed during the development of the University Admission Management System and maps the system features with standard software engineering practices.

4.1 Impact on Society, Environment and Sustainability

4.1.1 Impact on Life

This chapter describes how the University Admission Management System influences daily life and improves the admission process for students and administrators.

- **Time Saving:**
Students can apply online without visiting campus.
- **Convenience:**
Easy access from home using computer or mobile.
- **Less Paperwork:**
Reduces use of physical documents.
- **Better Accuracy:**
Minimizes human errors in data entry.
- **Transparency:**
Students can track application status easily.
- **Easy Management:**
Admin can manage large number of applications efficiently.

4.1.2 Impact on Society & Environment

This chapter explains how the University Admission Management System affects society and the environment. It highlights the social benefits and environmental improvements achieved through digitalization.

- **Easy Access for All:**

Students from different areas can apply online without difficulty.

- **Time & Cost Saving:**

Reduces travel expenses and saves time.

- **Transparency & Fairness:**

Ensures a clear and unbiased admission process.

- **Digital Awareness:**

Promotes use of technology in education.

- **Efficient Management:**

Helps universities handle large numbers of applications easily.

- **Less Paper Usage:**

Reduces physical forms and documents.

4.1.3 Ethical Aspects

This chapter discusses the ethical considerations involved in the University Admission Management System. It ensures that the system is used in a fair, secure, and responsible manner.

- **Data Privacy:**

Student personal information must be kept secure and confidential.

- **Security:**

Proper login system to prevent unauthorized access.

- **Fairness:**

Admission process should be unbiased and based on merit.

- **Transparency:**

Clear rules and visible admission status for all applicants.

- **Data Integrity:**

Information should not be altered or misused.

- **User Responsibility:**

Users should provide correct and honest information.

4.1.4 Sustainability Plan

The system uses a digital platform to reduce paper usage and support an eco-friendly environment. It can be easily updated and maintained for long-term use. The system is scalable, allowing it to handle increasing users over time. Regular maintenance ensures reliability, security, and continuous improvement.

4.2 Project Management and Team Work

With 5 members, we distributed the workload based on core technical and administrative tasks:

1. **Project Manager:** Responsible for documentation, timeline management, and final report compilation.
2. **Lead Developer :** Focused on the core C logic, including **Doubly Linked List** implementation and memory management.
3. **Database Designer :** Designed the data structures (`struct Student, struct AdmissionForm`) and handled data integrity.
4. **UI/UX Designer :** Designed the console-based interface, color schemes (`system ("COLOR 60")`), and menu flow.

5. **Quality Assurance/Tester** : Conducted bug testing, verified GPA logic, and ensured the admin password security works.

Cost Analysis & Budget

Item	Cost (BDT)
Development Cost	5,000
Software Tools	0 (Free)
Testing & Maintenance	2,000
Miscellaneous	1,000
Total Budget	8,000 BDT

Financial Analysis

S.N.	Component	Estimated Cost (BDT)
1	Internet & Electricity 	800
2	Report Printing & Binding 	1200
3	Stakeholder Visiting 	1500
4	Project Presentation Materials 	500
Total Estimated Cost		4000 

4.3 Complex Engineering Problem

4.3.1 Mapping of Program Outcome

In this section, provide a mapping of the problem and provided solution with targeted Program Outcomes (PO's).

Table 4.1: Justification of Program Outcomes

PO's	Justification
PO1	Justification of PO1 attainment
PO2	Justification of PO2 attainment
PO3	Justification of PO3 attainment

4.3.2 Complex Problem Solving

In this section, provide a mapping with problem solving categories. For each mapping add subsections to put rationale (Use Table 4.2). For P1, you need to put another mapping with Chapter 4. Engineering Standards and Mapping

4.3. Complex Engineering Problem

Knowledge profile and rational thereof.

Table 4.2: Mapping with complex problem solving.

EP1 Dept of Knowledge	EP2 Range of Conflicting Requirements	EP3 Depth of Analysis	EP4 Familiarity of Issues	EP5 Extent of Applicable Codes	EP6 Extent of Stake- holder Involvement	EP7 Interdependence
√	√				√	

EP1: Depth of Knowledge Required (√attained)

- The system requires advanced C programming for **Doubly Linked List** management.
- It utilizes dynamic memory allocation (`malloc`) for efficient real-time data storage.
- This goes beyond basic programming by applying complex data structure principles.

EP2: Range of Conflicting Requirements (√attained)

- The system must ensure **high security** for passwords while maintaining ease of use.
- It balances strict **GPA merit rules** against a limited number of departmental seats.
- Automated logic must resolve data integrity while handling high volumes of inputs.

EP6: Extent of Stakeholder Involvement (√attained)

- **Students** require a transparent application process and real-time status updates.
- **Admins** need robust control for merit list generation and document verification.
- **Accounts** rely on the system to track and verify student Transaction IDs.

4.3.3 Engineering Activities

In this section, provide a mapping with engineering activities. For each mapping add subsections to put rationale (Use Table 4.3).

Table 4.3: Mapping with complex engineering activities.

EA1 Range of re- sources	EA2 Level of Interaction	EA3 Innovation	EA4 Consequences for society and environment	EA5 Familiarity
√	√			

EA1: Range of Resources

- The project integrates advanced C-programming features like **Doubly Linked Lists** and dynamic memory management to handle student data.
- It utilizes a wide range of institutional resources, including departmental GPA requirements, faculty statistics, and financial datasets.

EA2: Level of Interaction

- The system facilitates high-level interaction between **Student** and **Admin** roles, ensuring real-time data updates across the platform.

- There is a seamless exchange between the GPA verification logic and the central database to automate the merit list generation process.

Chapter 5

Conclusion

This chapter summarizes the overall work and findings of the University Admission Management System. It highlights the system's achievements, benefits, and its impact on improving the admission process.

5.1 Summary

A University Admission Management System is software that automates and manages university operations like student enrollment, faculty details, courses, exams, fees, and library services. It improves efficiency, reduces paperwork, ensures data accuracy, and helps in quick decision-making. It benefits students, staff, and administration by centralizing all academic and administrative tasks.

5.2 Limitation

- High initial cost and maintenance.
- Requires technical skills to operate.
- Dependent on internet and electricity.
- May face data security and privacy issues.
- Limited flexibility for unique university policies.

5.3 Future Work

- **Student & Alumni Portal** – Enable career services, job placements, and alumni networking features.
- **Feedback & Survey System** – Collect feedback from students and staff for continuous improvement.
- **Resource Management** – Optimize classrooms, labs, and equipment scheduling using smart algorithms.
 - **E-Learning & Virtual Classrooms** – Add features for online lectures, assignments, and collaborative learning.
- **Automated Attendance & Exam Monitoring** – Use biometrics or smart devices to track attendance and exam integrity.
- **Data Analytics & Reporting** – Generate real-time reports on academic performance, finances, and operational efficiency.

References

1. S. Kaur, "*University Management System Using Software Engineering Approach*", International Journal of Computer Applications, 2015.
2. R. Sharma and P. Gupta, "*Design and Implementation of University Management System*", International Journal of Advanced Research in Computer Science, 2017.
3. R. S. Pressman, *Software Engineering: A Practitioner's Approach*, 8th Edition, McGraw-Hill, 2014.
4. <https://www.geeksforgeeks.org/university-management-system-project-in-java/>

ANNEXURE
DAFFODIL INTERNATIONAL UNIVERSITY
Daffodil Smart City, Ashulia, Dhaka, Bangladesh

Date: 01 April 2026

To,
The Manager / Coordinator
Daffodil Admission Office
Ashulia, Dhaka, Bangladesh

Subject: Request for Meeting and Information Sharing for Academic Project on University Admission Management System

Dear Sir/Madam,

We, the students of the Department of Computer Science and Engineering at Daffodil International University, are currently developing an academic project titled "University Admission Management System".

As your esteemed institution is widely recognized for its excellence in education and administration, we would like to request an opportunity to meet with your respected team to gain insights into your admission process.

We are particularly interested in understanding various aspects of the admission system, including student application procedures, eligibility verification, document submission, merit list preparation, admission confirmation, fee payment system, and the role of technology in managing these processes efficiently.

This interaction will significantly assist us in gathering practical requirements for designing a comprehensive, efficient, and user-friendly admission management system based on real-life workflows.

We assure you that we will respect your valuable time and approach the meeting with proper preparation. Any information shared with us will be used strictly for academic purposes only, and we will not request or utilize any confidential or sensitive data.

We would also be grateful if you could kindly provide us with a letter of acknowledgement or meeting confirmation, which is required for our academic project documentation.

Reference:

Mr. Raja Tariqul Hasan Tushar

Assistant Professor

Department of Computer Science and Engineering

Daffodil International University



Thank you for your kind consideration.

Sincerely yours,

Mahtab Hasan – ID: 252-15-860

Sumaiya Zamam Anika – ID: 252-15-874

Sneha Hajra – ID: 252-15-907

Ranggon Saha Riti – ID: 252-15-787

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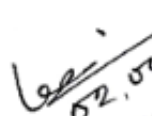
Signature: _____

Contact Information:

Email: 252-15-860@diu.edu.bd

Phone: 01736520402

Department of Computer Science and Engineering
Daffodil International University


02.04.2016
MD. LAZMINUR ALAM
Sr. Assistant Director
(Admission & Counselling)
Daffodil International University

